# 1 Lorentz transformations

1. Show that the wave equation

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial y^2} - \frac{\partial^2}{\partial z^2}\right)\phi(t,x,y,z) = 0$$

is *not* invariant under Galilean transformations, but *is* invariant under Lorentz transformations.

2. Show that all linear transformations

$$\begin{pmatrix} ct \\ x \end{pmatrix} = M \cdot \begin{pmatrix} ct' \\ x' \end{pmatrix}$$

which preserve the interval $(ct)^2 - x^2$ are given either by the matrix $M$ of the form

$$M = \Lambda = \begin{pmatrix} a & b \\ b & a \end{pmatrix},$$

where $a^2 - b^2 = 1$, or by the matrix $M$ which is the product of $\Lambda$ and the matrix

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

3. Consider two boosts in the same direction with the rapidity parameters $\theta_1$ and $\theta_2$. Show that they are equivalent to a single boost with the rapidity parameter $\theta = \theta_1 + \theta_2$. What is the rapidity parameter of the transformation inverse to the one with rapidity $\theta$?

4. Let an event have coordinates $x = 3\text{km}$, $t = 1\mu\text{s}$ in a frame $S$. What are the coordinates of the same event in a reference frame $S'$ that is moving with the velocity $\vec{v} = (V, 0, 0)$ relative to $S$? In which frame does this event happen at $t' = 0$? Draw a space-time diagram.

5. An astronaut on board of a spaceship of length $L = 300\text{m}$ has synchronized the rear and front lights so that they flash simultaneously. The spaceship moves with respect to a station with the velocity $v = 10^{-3}c$. When the spaceship passes near the station the lights flash. Would an observer on the station agree that the lights are synchronized? How big is the desynchronization from his point of view?

6. Two lights separated by a distance of 1km flash with a difference of $1\mu\text{s}$ in their rest frame. What is the velocity of the frame in which the flashes were emitted simultaneously ? Draw a space-time diagram.

7. Let the inertial frame $S'$ move with velocity $v$ relative to another inertial frame $S$. Let a point particle be measured at the space-time points $(ct, x)$ and $(c(t + dt), x + dx)$ in $S$. Compute these events in the system $S'$. Assume that the time $dt'$ that has elapsed in the system $S'$ is equal to the time $dt$ in $S$. Deduce that the velocity $u$ of the particle in $S$ is constant and compute its value. What is the value $u'$ of the velocity in $S'$? As the velocity of the particle is constant, the rest frame of the particle is also an inertial frame. Make a Lorentz transformation to that frame and compute the velocities of the two frames $S$ and $S'$ there. Use this to explain the equality of clocks $dt = dt'$.

8. $\mu$-mesons are unstable elementary particles whose lifetime is approximately $\tau \simeq 2 \cdot 10^{-6}$s. They are created in the atmosphere at the height of $\sim$ 10km by cosmic rays. Which velocity should $\mu$-mesons have in order to reach the ground before they decay?

9. A docking station for space ships consists of a cylinder of the length $L = 100$m with super-fast gates at both ends. The gates are synchronized to close and open simultaneously. A space ship of the length $l = 120$m is flying through the station at a speed $v$ close to the speed of light. How fast does the ship have to move for the gates to close and lock the ship inside the station (for a brief moment of time)? Consider the same events from the point of view of the ship crew.

10. The space ship crew from the previous problem decides to stop at the station to buy cigarettes. They start breaking when, according to the station observer, the ship is completely inside the station and the gates are closed. They manage to stop the ship before they reach the exit gates. But now the ship does not move and its length is larger than the station. How does it fit inside then?

11. Let the frame $S'$ move with velocity $\vec{V}_1 = (0, V, 0)$ with respect to $S$, while the frame $S''$ moves with velocity $\vec{V}_2 = (V, 0, 0)$ with respect to $S'$. Let the particle velocity in the frame $S''$ be $\vec{v}'' = (0, v, 0)$. Find the velocity of the particle in the frame $S$.

12. Let the frames $S$ and $S'$ be related by the boost with the velocity $V$ in the $x$-direction. In the frame $S'$ a particle is moving with the velocity $v'$ which forms the angle $\theta'$ with the $x$-axis. In the frame $S$ the angle between the particle velocity and the $x$-axis is $\theta$. Express $\theta$ in terms of the velocity $v'$, the angle $\theta'$ and the velocity $V$ between the frames.

13. In the previous exercise replace the particle by the light ray. Find the change in the observation angle $\Delta\theta = \theta' - \theta$ (the aberration angle) in the limit $V/c \ll 1$.

14. Consider two inertial frames $S$ and $S'$ with relative velocity $v$ and such that the origins coincide for $t = t' = 0$. After a time $\Delta t$ an observer at the origin of $S$ sends out a light signal which is received at time $\Delta t'$ by an observer at the origin of $S'$.

    (a) Suppose first that Galilean relativity holds, i.e. the two frames are related by a Galilean transformation. Compute $\Delta t'$ as a function of $\Delta t$ and $v$. (This is the non-relativistic Doppler shift for a receding observer.)

    (b) Deduce now the special relativistic Doppler shift by using Lorentz transformations. The result should be
    $$\Delta t' = \Delta t \sqrt{\frac{1 + v/c}{1 - v/c}}.$$
    Draw a space-time diagram of the situation.

## Quick answers

4. $V = 0.1c$
5. $\Delta t' = 10^{-9}$ s
6. $V = 0.3c$
8. $V = 0.998c$
9. $V = 0.55c$
11. $\vec{v} = \frac{1}{\gamma + Vv/c^2}(V, \gamma V + v, 0)$
12. $\theta = \arctan\left(\frac{v' \sin\theta'}{\gamma(v' \cos\theta' + V)}\right)$
13. $\Delta\theta \simeq \frac{V}{c} \sin\theta'$

# 2 Formalism of 4-vectors

1. Making use of the fact that the interval $s^2 = x_\mu x^\mu = (xx)$ is invariant under Lorentz transformations, find the matrix $A$ by which the *covariant* vectors are transformed,

$$x_\mu = A_\mu{}^\nu x'_\nu.$$

   Express $A$ in terms of the Lorentz transformation matrix $\Lambda^\mu{}_\nu$. Find $A$ explicitly in the case of the boost in the direction $x$ with the rapidity $\theta$.

2. Let $x^\mu$ and $y^\mu$ be two Lorentz vectors. Demonstrate that $x_\mu y^\mu = x^\mu y_\mu$.

3. Let $x^\mu$ be a time-like vector. Find a Lorentz transformation such that in the new frame the space components of the vector $x^\mu$ are zero[1].

4. Let $x^\mu$ be a space-like vector. Find a Lorentz transformation such that in the new frame the time component of the vector $x^\mu$ is zero.

5. Let $x^\mu_{(1)}$ be a time-like vector. Construct three vectors such that the resulting four vectors are linearly-independent and mutually orthogonal in the usual sense $(x_{(i)}x_{(j)}) = 0$ at $i \neq j$ where $i, j = 1, 2, 3, 4$. Are they time/space/light-like? Is there a time-like vector orthogonal to $x^\mu_{(1)}$?

6. Let $x^\mu$ be a null vector. Is it possible to find three more linearly independent vectors satisfying the orthogonality properties of the preceding exercise?

7. Let $x^\mu$ be a null vector. Find a linearly independent null vector.

8. Let $x^\mu$ and $y^\mu$ be a pair of linearly-independent null vectors. Construct two more linearly-independent vectors which are orthogonal to $x^\mu$ and $y^\mu$ and to each other.

9. Let $\Lambda^\mu{}_\nu$ be the matrix of the boost in the $x$-direction. Find the inverse matrix. Interpret the result.

10. Demonstrate that $\epsilon_{0123} = -1$.

11. Write out the differential operator $\partial^\mu \partial_\mu$ in components.

12. Find a Lorentz transformation that reverses time ($t' = -t$) and leaves the spatial directions invariant. Does this transformation preserve $\epsilon^{\mu\nu\rho\sigma}$? Is it orthochronous ?

13. Manipulating 4-vectors.

    (a) Let $x^\mu$ be a Lorentz vector. Demonstrate that $\epsilon^{\mu\nu\rho\sigma} x_\mu x_\sigma = 0$.

    (b) Demonstrate that $x^\nu y_\lambda \eta^{\rho\lambda} x^\mu y_\rho \eta_{\mu\nu} = x^2 y^2$. How does this quantity transform under a Lorentz boost ?

    (c) Demonstrate that $\eta^{\kappa\sigma} y_\mu \eta^{\rho\mu} x_\kappa y_\sigma x_\rho = (x \cdot y)^2$. How does this quantity transform under a Lorentz boost ?

    (d) Demonstrate that $\eta^\mu_\nu = \delta^\mu_\nu$.

    (e) Let $M^\mu_\nu$ be any matrix. How does $M^\mu_\mu$ transform under a Lorentz boost ?

[1] In this and a few subsequent exercises choose the orientation of the spatial axes so as to simplify calculations.

# 3 Relativistic energy and momentum

*Questions marked with an asterisk (*) are more difficult and not mandatory.*

1. Photons are particles of zero mass. Find the transformation rule of the photon energy and momentum under a boost with the velocity $V$ parallel to the photon momentum. Express the result in terms of the rapidity parameter.

2. The mass of the deuterium atom (D) is 1.8756 GeV ($1\text{GeV} = 1.602 \cdot 10^{-10}\text{J}$), while the mass of the helium atom ($^4$He) is 3.7284 GeV. Find the energy released in the conversion of 1 g of deuterium into helium (nuclear fusion).

3. A particle of mass $M$ moving with the velocity $V$ decays into two particles of masses $m_1$ and $m_2$. Find the relation between the energies of the resulting particles and the directions of their velocities.

4. Consider the elastic scattering of two particles of masses $m_1$ and $m_2$ from the center-of-momentum frame. Let $\vec{p}_1$, $\vec{p}_2$ be the three-momenta before the scattering and $\vec{p}'_1$, $\vec{p}'_2$ the three-momenta after the scattering process. Show that all momenta lie on a circle, i.e. are coplanar and of equal norm.

5. A moving particle of mass $m_1$ collides with the particle at rest of mass $m_2$. After the collision the velocity of the particle 1 changes the direction by the angle $\theta$. Demonstrate that in the case $m_1 > m_2$ the angle $\theta$ cannot exceed some maximum value. Find this value. Do the same calculation in Galilean relativity and comment on the result.

6. Consider the collision of the preceding question when the incident particle is massless. Find the relation between the energy of the incident particle after collision and its scattering angle.

7. Consider the non-relativistic head-on collision of a light particle of mass $m$ with a heavy one of mass $M$ at rest. Show that the energy transferred to the heavy particle, measured in units of the energy of the incident particle, goes to zero in the limit $m/M \to 0$. Demonstrate that this is not the case in the relativistic case.

8. Protons present in cosmic rays may interact with the photons of the cosmic microwave background. Consider a head-on collision (zero impact parameter) of a proton with a photon of the energy $2.6 \times 10^{-4}$eV. What minimum energy of the proton is needed to produce a $\pi^0$-meson in the reaction $p + \gamma \to p + \pi^0$ (the masses of particles are $m_p c^2 = 0.938\text{GeV}$, $m_\pi c^2 = 0.135\text{GeV}$, $m_\gamma = 0$)?

*9. A particle of mass $M$ which is at rest decays into three particles with masses $m_1$, $m_2$ and $m_3$. Find the maximum energy which can be carried away by one of these particles.

## Quick answers

2. $E = 3.42 \cdot 10^{21}\ \text{GeV} = 5.48 \cdot 10^{11}\ \text{J}$
3. $\cos\left(\widehat{P_i P_1}\right) = \frac{(m_2^2 - m_1^2 - M^2)c^4 + 2E_1(E_1+E_2)}{2\sqrt{E_1^2 - m_1^2c^4}\sqrt{(E_1+E_2)^2 - M^2c^4}}$
5. $|\sin\theta_{max}| = m_2/m_1$
6. $\cos\theta = 1 + m_2 c^2 \left(\frac{E'_1 - E_1}{E_1 E'_1}\right)$

# 4 Electrostatics

1. Prove the following relations:

$$\begin{aligned}
\vec{\nabla}(\vec{a}\cdot\vec{b}) &= (\vec{a}\cdot\vec{\nabla})\vec{b} + (\vec{b}\cdot\vec{\nabla})\vec{a} + \vec{a}\times(\vec{\nabla}\times\vec{b}) + \vec{b}\times(\vec{\nabla}\times\vec{a}) \\
\vec{\nabla}\cdot(\vec{a}\times\vec{b}) &= \vec{b}\cdot(\vec{\nabla}\times\vec{a}) - \vec{a}\cdot(\vec{\nabla}\times\vec{b}) \\
\vec{\nabla}\times(\vec{a}\times\vec{b}) &= \vec{a}(\vec{\nabla}\cdot\vec{b}) - \vec{b}(\vec{\nabla}\cdot\vec{a}) + (\vec{b}\cdot\vec{\nabla})\vec{a} - (\vec{a}\cdot\vec{\nabla})\vec{b}
\end{aligned}$$

2. Prove that for any scalar function $\phi(x,y,z)$ the relation

$$\vec{\nabla}\times(\vec{\nabla}\phi) = 0$$

holds. Show that for any vector field $\vec{A}(x,y,z)$ the following relation is true:

$$\vec{\nabla}\cdot(\vec{\nabla}\times\vec{A}) = 0.$$

3. Compute the Laplacian $\Delta = \vec{\nabla}\cdot\vec{\nabla}$ on a scalar function $\phi$ in spherical coordinates.

4. Consider the following two vector fields

   i) $\vec{A}(x,y,z) = A_0(yz + yz^2,\ xz + xz^2,\ xy + 2xyz)$

   ii) $\vec{B}(x,y,z) = B_0(x^2z,\ y^2z,\ xy^2)$

   where $A_0$ and $B_0$ are arbitrary (dimensionful) constants. Which of these vector fields can represent an electric field? For that field compute the resulting charge distribution and find a suitable electric potential.

5. Let the electric field be parallel to the $x$-axis in a given region of space,

$$\vec{E} = (E(x,y,z), 0, 0).$$

   a) Show that in that region the field cannot depend on $y$ and $z$.

   b) Show that if there are no charges in this region, the field is independent of $x$ as well.

6. Calculate the electric field and potential produced by a homogeneous charge distributed uniformly with the linear density $\lambda$ along the $z$-axis.

7. Calculate the electric field and potential produced by the following charge distributions:

   a) $\rho(r) = \rho_0$ for $r < R$ and zero otherwise

   b) $\rho(r) = C/r$ for $r < R$ and zero otherwise

   c) spherical shell of the radius $R$ and the surface charge density $\sigma$.

8. A perfect conductor has a cavity. Show that if a point charge $q$ is placed inside the cavity, the charge $-q$ is induced on the internal surface of the cavity.

9. A uniformly charged disk of the radius $a$ is placed at the origin in the plane $xy$. Calculate the electric field and potential produced by this disk at the points $(0,0,z)$ for positive and negative $z$. Calculate the limits $z \to 0_+$ and $z \to 0_-$.

10. Calculate the electric potential produced by two charges $Q$ and $Q'$, placed at $(0,0,d)$ and $(0,0,d')$, respectively. Find the equation of the surface of zero potential. Show that this surface is a sphere.

11. Show that the function
$$D(x,\alpha) = \frac{1}{\alpha\sqrt{2\pi}} \exp\left(-\frac{x^2}{2\alpha^2}\right)$$
goes to the Dirac $\delta$-function in the limit $\alpha \to 0$.

12. By using the Dirac $\delta$-function in appropriate coordinates, express the following charge distributions as three-dimensional charge densities $\rho(x)$:

    a) In spherical coordinates, a charge $Q$ uniformly distributed over a spherical shell of radius $R$.

    b) In cylindrical coordinates, a charge $\lambda$ per unit length uniformly distributed over a cylindrical surface of radius $b$.

    c) In cylindrical coordinates, a charge $Q$ spread uniformly over a flat circular disc of negligible thickness and radius $R$.

    d) The same as part c), but using spherical coordinates.

13. Each of three charged spheres of radius $R$, one conducting, one having a uniform charge density within its volume, and one having a spherically symmetric charge distribution that varies with radius as $r^n$ $(n > -3)$, has a total charge $Q$. Use Gauss's theorem to obtain the electric fields both inside and outside each sphere. Sketch the behavior of the field as a function of radius for the first two spheres, and for the third with $n = +2, -2$.

14. The time-averaged potential of a neutral hydrogen atom is given by
$$\Phi(r) = \frac{q}{4\pi\epsilon_0}\frac{\mathrm{e}^{-\alpha r}}{r}\left(1 + \frac{\alpha r}{2}\right),$$
where $q$ is the magnitude of the electron charge, and $\alpha^{-1} = a_0/2$, $a_0 = 0.529 \times 10^{-10}$ m being the Bohr radius. Find the distribution of charge (both continuous and discrete) that will give this potential and interpret the result physically.

15. Calculate the electrostatic potential produced by three point charges: $-Q$ in $(0,0,d)$, $2Q$ in $(0,0,0)$ and $-Q$ in $(0,0,-d)$. Calculate the multipole expansion up to and including the quadrupole term. Compare the exact result with the first non-zero term of the multipole expansion.

16. Calculate the electrostatic energy of the following charge distributions:

    (a) $\sigma = \frac{Q}{4\pi R^2}$ on the sphere of radius $R$

    (b) $\rho(r) = \begin{cases} \frac{3Q}{4\pi R^3} & 0 \le r \le R \\ 0 & r > R \end{cases}$

    If one describes the electron as the uniformly charged sphere (case (b)), one may derive the classical "electron radius" $r_e$ by equating the electrostatic energy to the rest energy of the electron $m_e c^2$. What is the value of $r_e$ thus obtained?

17. Calculate everywhere in space the solution to the Laplace equation that takes given values $\phi_a$ and $\phi_b$ on two concentric spheres of radii $a$ and $b$, $b > a$, and is zero at infinity.

18. Calculate everywhere in space the solution to the Laplace equation that takes given values $\phi_a$ and $\phi_b$ on two coaxial cylindrical shells (of infinite heights) of radii $a$ and $b$, $b > a$. What is the condition on $\phi_b$ for the value of the potential to be 0 at infinity ?

19. Let a conducting sphere $S_1$ of radius $a$ carrying charge $Q$ be placed in the middle of a spherical cavity $S_2$ of radius $b$ in a perfect conductor held at zero potential. Find the potential on the sphere $S_1$.

20. Let a charge $Q$ be placed at the point $(0, 0, d)$ above a conducting plane $z = 0$ held at zero potential. a) Find the potential everywhere in space. b) Calculate the density of charge and the total charge induced on the plane. c) Calculate the force acting on the charge $Q$. d) Calculate the work needed to bring this charge from infinity to the point $(0, 0, d)$.

21. Let $Q$ be a point charge placed outside of a conducting sphere of radius $a$ held at zero potential. a) Calculate the distribution of charge induced on the sphere. b) Calculate the total charge of the sphere. The same questions for the charge placed inside the sphere.

22. Calculate the potential produced by a point charge $Q$ placed at a distance $R$ from the center of a conducting sphere of radius $a$ having total charge $q$.

23. Consider the analog of the Poisson equation in the space with different number of dimensions $n = 1, 2, 4, ...,$

$$\left(\frac{\partial^2}{x_1^2} + \frac{\partial^2}{x_2^2} + \ldots + \frac{\partial^2}{x_n^2}\right)\Phi \equiv \Delta^{(n)}\Phi = \rho(x).$$

Find the Green's function of the corresponding Laplace equation, i.e., the function $G(x, x')$ satisfying the equation

$$\Delta^{(n)}G(x, x') = \delta^n(x - x').$$

24. Calculate the electrostatic potential inside a rectangular box of dimensions $0 \le x \le a$, $0 \le y \le b$, $0 \le z \le c$, if all sides of the box are held at zero potential except the one at $z = c$ where the potential is $V(x, y) = V_0 \sin(\pi x/a) \sin^3(\pi y/b)$.

25. Calculate the electrostatic potential and electric field produced by an infinite cylinder of radius $a$ with the axis along the direction of the axis $z$ held at zero potential, placed in an external uniform field $\vec{E} = (E, 0, 0)$. *Hint: use the cylindrical coordinates; outside of the cylinder solve the Laplace equation by separation of variables.*

## Quick answers

3. $\Delta f = \frac{1}{r}\frac{\partial^2}{\partial r^2}(rf) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial f}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial\varphi^2}$
4. $\vec{B}$ does not work, $\vec{A}$ works with $\rho = 2A_0\epsilon_0 xy$, $\phi = -A_0(xyz + xyz^2)$
6. $\vec{E} = \lambda\vec{u}_r/2\pi r\epsilon_0$, $\phi = -\lambda\ln(r)/2\pi\epsilon_0$
9. $\vec{E}(0, 0, z) = \sigma/2\epsilon_0 \cdot \left(\pm 1 - \frac{z}{\sqrt{a^2+z^2}}\right)\vec{u}_z$ for $z \pm 0$, $\phi = \sigma/2\epsilon_0 \cdot \left(\sqrt{a^2+z^2} \mp z\right)$ for $z \pm 0$

14. Continuous ($r > a_0$): $\rho(r) = -q\alpha^3 e^{-\alpha r}/8\pi$, discrete ($r << a_0$): $\rho(r) = q\delta(r)$
16. $r_e = 1.7 \cdot 10^{-15}\text{m}$
19. $\phi(a) = Q/4\pi\epsilon_0 \cdot (1/a - 1/b)$

# 5 Magnetostatics

1. Calculate everywhere in space the magnetic field produced by a coaxial cable composed of a conducting cylinder or radius $a_1$, and a hollow cylinder of internal radius $a_2 > a_1$ and external radius $a_3$. The internal and external cylinders carry total currents $I$ and $-I$, respectively. Both currents are uniformly distributed over the section of the conductors.

2. Calculate at the point $(0, 0, z)$ the magnetic field of a circular loop of radius $a$ centered around the origin in the $(x, y)$ plane. Consider the limit $z \gg a$. Calculate the leading contribution to the magnetic field at $|\vec{x}| \gg a$ everywhere in space. Compare to the electric field of a dipole.

3. A uniformly charged sphere with charge density $\rho$ centered at the origin is rotating around the $z$-axis with constant angular velocity $\omega$. Find the resulting current density $\vec{J}(x)$.

4. A uniformly charged disk of radius $a$ centered at the origin in the plane $z = 0$ is rotating around $z$-axis with constant angular velocity $\omega$. Find the magnetic field at points of the $z$-axis. Find the leading contribution to the field at $|\vec{x}| \gg a$.

5. Consider the magnetic field $\vec{B} = (0, 0, B_z)$ or constant $B_z$. Give several vector potentials $\vec{A}$ which give rise to this magnetic field. Check that they differ by a gauge transformation $\vec{A} \to \vec{A} + \vec{\nabla}\Psi$.
   Consider now a general constant magnetic field $\vec{B}$. Can the vector potential be given by
   $$\vec{A} = \frac{1}{2}\vec{B} \times \vec{x}\,?$$

6. Let $S$ be a piece of a 2-dimensional surface with boundary $\partial S$ situated in a region of space filled with a static non-uniform magnetic field $\vec{B}(\vec{x})$. Demonstrate the relation
   $$\int_S d\vec{s} \cdot \vec{B} = \int_{\partial S} d\vec{l} \cdot \vec{A}.$$
   In other words, the flux of the magnetic field through any surface is equal to the line integral of the vector potential along the boundary of this surface.

7. Consider a solenoïde of radius $a$ with its axis directed along the $z$-axis. Let the constant magnetic field inside the solenoïde be $\vec{B} = (0, 0, B_z)$, while the magnetic field outside be zero, $\vec{B}_{\text{out}} = 0$. Does the vector potential vanish outside of the solenoïde? Can it be represented as the gradient of a scalar function? Can the following expression represent the potential outside of the solenoïde,
   $$\vec{A} = \frac{C}{x^2 + y^2}(-y, x, 0).$$
   Calculate the integral of $\vec{A}$ along the circular contour going around the solenoïde outside of it and relate the constant $C$ to the flux of the magnetic field.

8. A square conducting frame with sides of length $a$ carries a current $I$. The frame is placed in a uniform magnetic field $\vec{B}$ in such a way that one pair of opposite sides is perpendicular to the magnetic field. Calculate the moment of force (torque, *couple*) acting on the frame for different orientations of the frame with respect to the magnetic field. Express the result in terms of the magnetic moment of the frame.

9. Two coaxial hollow cylinders of radii $a_1$ and $a_2 > a_1$ carry total currents $I$ and $-I$, respectively. The currents are uniformly distributed over the surfaces of the cylinders. Calculate the energy, per unit length of the cylinders, contained in the magnetic field produced by these currents.

10. A square conducting frame with the side $a$ is placed in a uniform magnetic field $\vec{B}$ perpendicular to the field. The frame starts to rotate with an angular velocity $\vec{\omega}$ perpendicular to $\vec{B}$. Find the electromotive force *emf* induced in the frame as a function of time. Does it depend on the position and orientation of the rotation axis?

## Quick answers

1. $$\begin{cases} B &= (\mu_0 I r)/(2\pi a_1^2) \text{ for } r < a_1 \\ B &= (\mu_0 I)/(2\pi r) \text{ for } a_1 \leq r < a_2 \\ B &= (\mu_0 I)/(2\pi r) \cdot (1 - (r^2 - a_2^2)/(a_3^2 - a_2^2)) \text{ for } a_2 \leq r < a_3 \\ B &= 0 \text{ for } r \geq a_3 \end{cases}$$

2. $\vec{B}(0,0,z) = \frac{\mu_0 I a^2}{2(a^2+z^2)^{3/2}} \vec{u}_z$
3. $\vec{J} = \rho r \omega \sin\theta \vec{u}_\varphi$
4. $\vec{B}(0,0,z) = \frac{\mu_0 \sigma \omega (z \pm \sqrt{a^2+z^2})^2}{2\sqrt{a^2+z^2}} \vec{u}_z$ for $z \mp 0$
8. $\vec{T} \simeq I a^2 B \cos\theta \vec{u}_z$ (depends on the chosen frame)
9. $\omega/h = \mu_0 I^2 \ln(a_2/a_1)/4\pi$
10. $\varepsilon = B w a^2 \sin(\omega t)$